

# EL Calls

Developer Guide — Architecture, Data Model, Edge Functions, Performance and Security

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## 1. Stack Overview

Frontend: React 18, Vite 5, TypeScript 5, Tailwind v3, shadcn/ui, TanStack Query, React Router v6.  
Backend: Lovable Cloud (managed Supabase) — Postgres 15 with RLS, Deno edge functions, Realtime broadcast, Storage. Voice: Twilio Programmable Voice + browser WebRTC. AI: Lovable AI Gateway (Gemini 2.5 Flash for chat/officer-chat, Whisper-equivalent for transcription).

### 1.1 Repository layout

```
src/  
components/ shadcn-based UI, feature panels under dashboard/  
hooks/ useAuth, useUserRole, useWebRTCcall, useAgentPresence...  
integrations/supabase/ generated client + types (DO NOT EDIT)  
pages/ route components  
lib/ phone, normalizers, utilities  
supabase/  
functions/ Deno edge functions (one folder = one function)  
migrations/ timestamped SQL files  
load-tests/ k6 scripts and reports  
public/docs/ User and Developer PDFs (this file)
```

## 2. Routing and Navigation

React Router v6 with a ScrollToTop component remounting on every pathname change. AdminRoute and ProtectedRoute wrap restricted routes and call useUserRole, which queries user\_roles via the has\_role RPC. Sidebar items are filtered by role on the client; server enforcement is the source of truth.

## 3. Data Model

### 3.1 Core tables

| Table      | Purpose                                                                         |
|------------|---------------------------------------------------------------------------------|
| profiles   | Public user info, linked to auth.users.                                         |
| user_roles | Role assignment (admin/supervisor/agent). Separate table to prevent escalation. |
| clients    | Customer master records with tags.                                              |

| Table                                            | Purpose                                                                  |
|--------------------------------------------------|--------------------------------------------------------------------------|
| call_campaigns / campaign_types                  | Campaign definitions and templates.                                      |
| campaign_runs / campaign_jobs                    | Per-execution and per-target row.                                        |
| outbound_calls                                   | One row per dialed call. 6 composite indexes added in 2026-06 migration. |
| outbound_call_events (+monthly partitions)       | Append-only event log from Twilio status callbacks.                      |
| call_queue / call_transfers                      | Live routing primitives.                                                 |
| agent_status / agent_skills / agent_shifts       | Presence, capabilities, rota.                                            |
| escalation_settings / emergency_alerts           | Escalation rules and raised alerts.                                      |
| call_transcriptions / campaign_recordings        | ASR output and recording metadata.                                       |
| ivr_menu_options / supported_languages           | IVR config and localisation.                                             |
| audit_log / error_events / system_health_metrics | Observability.                                                           |

## 3.2 Row Level Security

Every public table is RLS-enabled. The default pattern: enable RLS, GRANT to the appropriate roles, then create policies that delegate to security-definer helpers.

```
create or replace function public.has_role(_uid uuid, _role app_role)
returns boolean language sql stable security definer
set search_path = public as $$
select exists(select 1 from public.user_roles where user_id=_uid and role=_role)
$$;

create policy "Admins manage all" on public.call_campaigns
for all to authenticated using (public.has_role(auth.uid(), 'admin'));
```

Partitioned tables (outbound\_call\_events\_YYYY\_MM) inherit RLS from the parent and are explicitly RLS-enabled on creation by create\_outbound\_call\_events\_partition().

## 4. Edge Functions

All functions live under *supabase/functions/<name>*. Naming convention: kebab-case folder; index.ts as endpoint. Shared utilities in *\_shared/* include twilio-verify (signature validation), idempotency, rate-limit, audit, phone and monitor.

| Function                                    | Auth                 | Purpose                                                                                      |
|---------------------------------------------|----------------------|----------------------------------------------------------------------------------------------|
| ai-voice-call                               | Twilio signature     | Initial TwiML for outbound dialer.                                                           |
| ai-voice-call-status                        | Twilio signature     | Inserts every Twilio status into <code>outbound_call_events</code> (pure append, no UPDATE). |
| ai-voice-call-dtmf*                         | Twilio signature     | DTMF gather flows including appointment + language.                                          |
| ai-voice-call-bridge-ivr                    | Twilio signature     | Connects callers to the configured IVR menu.                                                 |
| process-call-events                         | CRON_SECRET          | Drains <code>outbound_call_events</code> into <code>outbound_calls</code> in batches of 500. |
| campaign-scheduler                          | Staff JWT            | Plans which campaigns to run within dialing windows.                                         |
| campaign-enqueue                            | Staff JWT            | Enqueues <code>campaign_jobs</code> into <code>call_queue</code> .                           |
| campaign-worker                             | CRON_SECRET          | Pops jobs and triggers Twilio dials.                                                         |
| campaign-control                            | Staff JWT            | Start/pause/resume campaigns.                                                                |
| retry-campaign-calls                        | CRON_SECRET or staff | Re-attempts failed jobs per retry policy.                                                    |
| reconcile-call-status                       | CRON_SECRET          | Hourly reconciliation against Twilio API.                                                    |
| smart-call-routing                          | Staff JWT            | Scores agents by skills+proficiency+performance.                                             |
| analyze-call-sentiment                      | Internal             | Lovable AI sentiment per transcript turn.                                                    |
| transcribe-call / voice-to-text             | Staff JWT, 10MB cap  | Whisper transcription.                                                                       |
| officer-chat / ai-policy-notes              | Authenticated        | Gemini 2.5 Flash assistants grounded in <code>knowledge_base</code> .                        |
| send-agent-invitation                       | Staff JWT            | Server forces <code>invitedBy</code> = caller identity.                                      |
| send-escalation-notification                | Staff JWT            | Pulls recipients from <code>escalation_settings</code> only.                                 |
| send-appointment-email / send-payment-email | Internal             | Transactional emails via SendGrid.                                                           |
| manage-agents / manage-app-secrets          | Admin JWT            | Admin CRUD.                                                                                  |
| twilio-recent-calls / twilio-verify         | Staff JWT            | Diagnostics.                                                                                 |

| Function                  | Auth          | Purpose                              |
|---------------------------|---------------|--------------------------------------|
| suggest-tag               | Staff JWT     | Heuristic + AI tag suggestions.      |
| sample-data               | Staff JWT     | Seed demo data.                      |
| gdpr-export / gdpr-delete | Admin JWT     | Privacy operations.                  |
| realtime-chat             | Authenticated | WebSocket relay for live chat panel. |

## 4.1 Error handling contract

Edge functions log full errors server-side and return a generic message to the client to avoid information disclosure:

```
try { /* work */ }
catch (e) {
  console.error('fn-name failed', e);
  return new Response(JSON.stringify({ error: 'An unexpected error occurred' })),
  { status: 500, headers: corsJson });
}
```

## 4.2 Twilio webhook security

Inbound Twilio callbacks (status, DTMF, IVR bridge) validate the X-Twilio-Signature header via `_shared/twilio-verify.ts` using the auth-token secret. Replay protection uses the idempotency helper keyed on CallSid+CallStatus.

## 5. Performance Architecture

### 5.1 Index strategy

Migration 2026-06-22 added six composite indexes on `outbound_calls`:

- `twilio_call_sid` (webhook lookups, O(1))
- `campaign_id + created_at DESC` (campaign dashboards)
- `call_status + created_at DESC` (live queues)
- `agent_email + created_at DESC` (agent performance)
- `phone_number` (fallback matching)
- `client_id + created_at DESC` (client-level queries)

### 5.2 Event-log + partitioning

`ai-voice-call-status` is hot during campaigns: 4-6 callbacks per call. The previous design did an `UPDATE` on `outbound_calls` per callback and contended on the same row. The new design:

- Webhook does pure `INSERT` into `outbound_call_events` (range-partitioned by month).
- `process-call-events` runs every 5-10s on a cron, batching 500 rows.
- It resolves call IDs via `twilio_call_sid` or `parent_sid` index, then applies the final state with a single `UPDATE` per call inside one transaction.
- `create_outbound_call_events_partition()` pre-creates next month's partition and enables RLS automatically.

```
-- drainer (simplified)
select public.process_outbound_call_events(500);
```

### 5.3 Load test results (k6)

| Scenario           | Load             | p95    | Throughput         | Notes                            |
|--------------------|------------------|--------|--------------------|----------------------------------|
| Campaign execution | 200 RPS, 5m      | 220 ms | 98.4% OK           | DB CPU 55%.                      |
| Concurrent dialing | 500 active calls | n/a    | Stable             | Twilio rate is the cap.          |
| Webhook processing | 1200 RPS         | 85 ms  | 100% OK (post-fix) | Was 22% errors before event-log. |
| Dashboard traffic  | 200 vUsers, 10m  | 180 ms | 99.9% OK           | Indexes cut p95 by 6x.           |

Max sustained capacity: ~130k calls/day per project on Lovable Cloud Pro plan; bottleneck shifts to Twilio concurrent-call quota above that.

## 6. Security

- Roles stored in `user_roles` table only; checked via `has_role` security-definer.

- RLS on every public table and every partition. GRANTS explicit, never relying on defaults.
- Edge functions classify auth: public + Twilio-signed, authenticated, staff JWT, admin JWT, CRON\_SECRET.
- No SUPABASE\_SERVICE\_ROLE\_KEY usage on the client; only edge functions read it.
- Generic error messages on all public surfaces; full errors logged via console.error.
- Rate-limiting helper in \_shared/rate-limit.ts protects expensive endpoints.
- Audit-log entries written for all role changes, GDPR ops, secret reads, escalations.
- Auth redirects pinned to https://elcalls.lovable.app to avoid open-redirect.
- Sign-out forces redirect to '/' and clears all client state.

## 7. Realtime and Presence

Supabase Realtime broadcast is used for: live transcription updates, queue and transfer changes, supervisor live view, and officer-chat streaming. useAgentPresence sends a heartbeat every 30s and flips users to away after 5 minutes of inactivity. WebRTC signalling rides the same Realtime channel.

## 8. WebRTC Calling

useWebRTCCall sets up RTCPeerConnection per call, with ICE servers fetched from a signed edge function. SDP offers/answers and ICE candidates broadcast through a Supabase Realtime channel scoped to the call ID. The LiveKit credentials slot exists for a future migration to LiveKit SFU when concurrent browser-to-browser calls exceed the mesh limit.

## 9. AI Integrations

- Officer Chat and AI Policy Notes use Lovable AI Gateway with google/gemini-2.5-flash by default.
- Tag suggestions use a heuristic pass plus an AI fallback.
- Sentiment runs per transcript turn and writes to `call_transcriptions.sentiment`.
- Transcription uses Whisper via voice-to-text/transcribe-call; 10MB payload cap.

All AI calls go through the gateway — no third-party API keys are required from the customer for chat/embeddings/STT/TTS unless they want a specific provider.

## 10. Local Development

```
bun install
bun run dev # vite on :8080
bunx vitest run # unit tests
bunx tsgo --noEmit # typecheck (preferred over tsc)

# edge functions: deploy via Lovable Cloud, logs in dashboard
# never edit src/integrations/supabase/{client,types}.ts (auto-gen)
```

## 11. Migrations

Every CREATE TABLE in *public* is followed in the same migration by GRANTs and RLS enable + policies. The repo includes a migration that backfills RLS on existing partitions and adds the `create_outbound_call_events_partition()` helper.

## 12. Observability

- `error_events` captures handled exceptions from edge functions.
- `system_health_metrics` records throughput, queue depth, ASR latency.
- `audit_log` records every sensitive action (role changes, secret reads, GDPR ops).
- Lovable Cloud function logs are the primary surface for live debugging.

## 13. Testing

- Unit tests: vitest, including `src/lib/phone.test.ts` and `supabase/functions/_shared/*.test.ts`.
- Load tests: k6 scripts under `load-tests/scripts` (01 campaign, 02 dialing, 03 webhook, 04 dashboard).
- Run all: `bunx vitest run`; load report at `load-tests/reports/REPORT.md`.

## 14. Deployment

Use the Publish action in Lovable. Custom domains map to the published URL. Cron schedules for process-call-events, campaign-worker, reconcile-call-status and retry-campaign-calls are configured in Lovable Cloud. CRON\_SECRET protects all cron-triggered endpoints.

## 15. Extending the Platform

- New campaign type: insert into campaign\_types, add any IVR menu options, deploy.
- New language: insert supported\_languages row, upload audio via LanguageAudioUploader, add translation strings.
- New role: extend app\_role enum, add policies, update AdminRoute logic.
- New AI feature: prefer Lovable AI Gateway; never embed third-party keys in client.

## 16. Coding Conventions

- Tailwind tokens only; no hardcoded colour utilities (no text-white, bg-[#hex]).
- Long forms broken into tabs so action buttons stay visible.
- Wow aesthetic: gradients, glassmorphism, micro-interactions (ripples, confetti).
- Officer chat always rendered as a floating non-modal right-side panel.
- Sidebar collapsible from the left; scroll to top on every nav.

## 17. Appendix: Useful RPCs

| RPC                                                | Returns | Used for                   |
|----------------------------------------------------|---------|----------------------------|
| has_role(uid, role)                                | boolean | RLS policies               |
| is_admin() / is_staff() / is_supervisor_or_admin() | boolean | Edge function gating       |
| process_outbound_call_events(limit)                | integer | Drainer cron               |
| create_outbound_call_events_partition(date )       | void    | Monthly partition creation |



## 18. Active Directory (SAML SSO) Setup

The Active Directory configurator (Setup ! People ! Active Directory) provides a guided, auto-validated workflow for connecting Care Connect to a corporate IdP. Implementation: `src/components/dashboard/ActiveDirectorySetup.tsx`.

### 18.1 Supported IdP Types

A discriminated `idpTemplates` map drives the dynamic field set per directory type: Microsoft Entra ID (Azure AD), AD FS, Okta, OneLogin, Google Workspace and Generic SAML 2.0. Selecting a type swaps the required fields and resets verification state.

### 18.2 Draft Save & Resume

Configuration is persisted as a draft in `localStorage` under key `el:ad-setup:draft:v1`. The draft preserves: selected IdP type, captured field values, allowed domains, pilot users, default role, enforcement toggles, break-glass email, and the verified metadata snapshot (when status is 'ok'). On dialog open the draft is hydrated and the user is notified. A 'Draft saved' badge plus a 'Resume Active Directory setup' CTA appear on the card when a draft exists. Submission clears the draft via `clearDraft()`.

### 18.3 Entra ID Consistency Validation

When the IdP type is entra, an `entraIssues` memo runs the following checks before any checklist item turns green:

- Tenant ID matches GUID regex `^[0-9a-f]{8}-...-[0-9a-f]{12}$`
- Enterprise App Object ID matches the same GUID regex
- Federation Metadata URL host is `login.microsoftonline.com` (or `.us`)
- Metadata URL contains the supplied Tenant ID
- Metadata URL path includes `/federationmetadata/`
- After fetch, the returned `entityID` references the Tenant ID

### 18.4 Test SSO Button

`runSsoTest()` performs a live probe against the IdP. Preconditions: all required fields supplied, metadata verified, and (for Entra) consistency checks passing. The probe fetches the SSO endpoint extracted from the verified metadata using mode: 'no-cors'; an opaque response counts as reachable. The result is stored in `ssoTest` state as `idle` | `running` | `success` | `error`, surfaced both inline and as a checklist row with a specific failure reason.

### 18.5 Auto-Verified Production Checklist

The checklist is computed by an `autoChecks` memo — users cannot tick items manually. Each entry exposes label, done, and a why string explaining the current state. Green rows show a confirmation; red rows show the precise blocker.

- All required IdP fields supplied — lists missing field labels when red.
- IdP details consistent — surfaces the first Entra consistency error.
- IdP metadata verified & reachable — reports HTTP/parse failure or asks for Verify click.
- Live SSO test passed — shows the probed hostname or the specific failure reason.
- Email domains allowlisted — lists malformed domains.
- Pilot users supplied (2-5 valid emails) — explains count or format violation.
- Single Logout + SSO enforcement enabled.
- Break-glass admin email captured (or explicitly disabled).

Submit is enabled only when every check is green. On submit the component dispatches a `lovable:saml-sso-submit` CustomEvent with the full payload and clears the draft. Server-side SAML trust finalisation is handled by Lovable Cloud SSO configuration.

### 18.6 Service Provider Constants

- Entity ID: `https://elcareconnect.lovable.app/saml/metadata`
- ACS / Reply URL: `https://elcareconnect.lovable.app/saml/acs`
- Single Logout URL: `https://elcareconnect.lovable.app/saml/slo`
- NameID format: `emailAddress`